THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET A

Chapters 1–4: **Chemical Reactions and Equations, Acids, Bases and Salts, Metals and Non-metals, Carbon and its Compounds** | Class 10 Science | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 12 × 1 = 12

Q.	Correct Option	Brief Reasoning
1	**(a)** double displacement (precipitation)	Exchange of ions; white $BaSO_4$ precipitate
2	**(c)** silver chloride	$AgNO_3 + NaCl \longrightarrow AgCl\downarrow + NaNO_3$
3	**(b)** NO_2	Nitrogen dioxide
4	**(b)** 5.5	Enamel corrodes below 5.5
5	**(d)** oxalic acid	NCERT table
6	**(c)** Incorrect — less reactive metals like copper do not give hydrogen	Cu does not displace H
7	**(a)** sodium	Catches fire in air
8	**(c)** silver and copper	Lead and mercury are poor
9	**(d)** bromine	Mercury is a metal
10	**(d)** fullerene	C_{60} buckminsterfullerene
11	**(d)** ethanoic acid	Two-carbon acid
12	**(b)** ethyne	Acetylene

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
13	It is covered with a layer of magnesium oxide/basic magnesium carbonate, which prevents it from burning; cleaning removes this layer.	1+1
14	It contains ions (H^+/H_3O^+ and anions) which carry current.	2
15	Calcium sulphate hemihydrate, $CaSO_4 \cdot \frac{1}{2}H_2O$	1+1
16	False — HNO_3 is a strong oxidising agent; it oxidises H_2 to water (except Mg and Mn with very dilute HNO_3).	2
17	Carbon has four valence electrons and forms four covalent bonds.	2

SECTION — C (Short Answer) — Answers — 6 × 3 = 18

Q.	Expected Answer / Value Points	Marks
18	Brown coating on the nail; blue solution turns light green. $Fe + CuSO_4 \longrightarrow FeSO_4 + Cu$ — displacement	1+1+1
19	Electrolysis of brine: $2\,NaCl(aq) + 2\,H_2O \longrightarrow 2\,NaOH + Cl_2 + H_2$; chlorine at anode, hydrogen at cathode, NaOH near cathode	1+1+1
20	Reaction of iron(III) oxide with aluminium: $Fe_2O_3 + 2\,Al \longrightarrow 2\,Fe + Al_2O_3$ + heat; used to join railway tracks/cracked machine parts	1+1+1
21	Unsaturated compounds add atoms across double/triple bond, e.g. $CH_2{=}CH_2 + H_2 \xrightarrow{Ni} C_2H_6$; hydrogenation of vegetable oils	1+1+1
22	(i) C–C single bond, 3 H each (ii) C=C double bond, 2 H each (iii) O=C=O with two double bonds and lone pairs on O	1+1+1
	OR: A series of compounds with the same functional group and similar properties; successive members differ by CH_2 (14 u); same general formula; gradual change in physical properties. e.g. CH_4, C_2H_6, C_3H_8	1+2
23	(i) $3\,H_2 + N_2 \longrightarrow 2\,NH_3$ (ii) $2\,H_2S + 3\,O_2 \longrightarrow 2\,H_2O + 2\,SO_2$ (iii) $3\,BaCl_2 + Al_2(SO_4)_3 \longrightarrow 3\,BaSO_4 + 2\,AlCl_3$	1+1+1
	OR: Paint prevents contact of iron with air and moisture, so it does not rust. Nitrogen replaces oxygen and prevents oxidation of fats/oils (rancidity).	1½+1½

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
24	(i) methanoic (formic) acid (ii) it is a mild base which neutralises the acid (iii) neutralisation (iv) more than 7 (v) calamine solution / dock plant leaves for nettle	1+1+1+1+1

	OR: (i) Heating gypsum at 373 K: $\mathrm{CaSO_4 \cdot 2\,H_2O \longrightarrow CaSO_4 \cdot \tfrac{1}{2}\,H_2O + \tfrac{3}{2}\,H_2O}$ (ii) sets into hard gypsum (iii) plaster for fractured bones, toys, decoration, smooth surfaces (iv) it absorbs moisture and sets hard	2+1+1+1
25	(i) $\mathrm{CH_3COOH + NaOH \longrightarrow CH_3COONa + H_2O}$; $\mathrm{2\,CH_3COOH + Na_2CO_3 \longrightarrow 2\,CH_3COONa + H_2O + CO_2}$; $\mathrm{CH_3COOH + C_2H_5OH \xrightarrow{acid} CH_3COOC_2H_5 + H_2O}$ (ii) it ionises partially	3+2
	OR: (i) n-pentane $\mathrm{CH_3(CH_2)_3CH_3}$, isopentane $\mathrm{CH_3CH(CH_3)CH_2CH_3}$, neopentane $\mathrm{C(CH_3)_4}$ (ii) same molecular formula, different structures (iii) no — only one carbon, a single arrangement	3+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET B

Chapters 1–4: **Chemical Reactions and Equations, Acids, Bases and Salts, Metals and Non-metals, Carbon and its Compounds** | Class 10 Science | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 12 × 1 = 12

Q.	Correct Option	Brief Reasoning
1	**(a)** gain of oxygen or loss of hydrogen	Definition
2	**(c)** iron displaces copper, forming $FeSO_4$	Fe is more reactive
3	**(b)** nitrogen	Inert, prevents oxidation of fats
4	**(b)** calcium oxychloride	$CaOCl_2$
5	**(b)** 7	e.g. NaCl is neutral
6	**(b)** Incorrect — only water-soluble bases are alkalis	e.g. $Cu(OH)_2$ is a base but insoluble
7	**(c)** zinc	Zn is more reactive
8	**(d)** high melting points	Strong electrostatic forces
9	**(c)** malleability	Gold and silver are most malleable
10	**(b)** hydrogenation	Addition reaction
11	**(a)** –OH	Alcohol
12	**(b)** C_5H_{10}	Starts from C_2H_4

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
13	False — it is photolytic decomposition: $2\,AgCl \xrightarrow{\text{light}} 2\,Ag + Cl_2$; the grey colour is due to silver.	1+1
14	False — they do not ionise to give H^+ ions in water, so they do not conduct electricity and are not acidic.	2
15	A homogeneous mixture of a metal with other metals or a non-metal; e.g. brass (Cu + Zn), bronze (Cu + Sn), solder (Pb + Sn)	1+1
16	It reacts vigorously with oxygen and moisture and can catch fire; kerosene prevents contact with air.	2
17	Gaining or losing 4 electrons needs too much energy, so carbon shares electrons.	2

SECTION — C (Short Answer) — Answers — 6 × 3 = 18

Q.	Expected Answer / Value Points	Marks
18	(i) Calcium hydroxide (slaked lime) (ii) combination, exothermic (iii) $Ca(OH)_2 + CO_2 \longrightarrow CaCO_3 + H_2O$ — a thin shiny layer of $CaCO_3$ forms	1+1+1
19	(i) D (ii) C (iii) B (iv) A (v) E. Increasing H^+: C < E < D < A < B	2+1
20	(i) $3\,Fe + 4\,H_2O \longrightarrow Fe_3O_4 + 4\,H_2$ (ii) $Ca + 2\,H_2O \longrightarrow Ca(OH)_2 + H_2$ (iii) $2\,K + 2\,H_2O \longrightarrow 2\,KOH + H_2$ + heat	1+1+1
21	Compounds with the same molecular formula but different structures; n-butane $CH_3-CH_2-CH_2-CH_3$ and isobutane $CH_3-CH(CH_3)-CH_3$	1+1+1
22	Green crystals turn white then reddish-brown; smell of burning sulphur. $2\,FeSO_4 \xrightarrow{\text{heat}} Fe_2O_3 + SO_2 + SO_3$ — thermal decomposition	1+1+1
	OR: (i) $BaCl_2\,(aq) + Na_2SO_4\,(aq) \longrightarrow BaSO_4\,(s) + 2\,NaCl\,(aq)$ (ii) $NaOH\,(aq) + HCl\,(aq) \longrightarrow NaCl\,(aq) + H_2O\,(l)$ (iii) $Zn\,(s) + 2\,AgNO_3\,(aq) \longrightarrow Zn(NO_3)_2\,(aq) + 2\,Ag\,(s)$	1+1+1
23	Baking soda + mild edible acid (tartaric acid). On heating, CO_2 is released which makes the cake rise. With only baking soda, sodium carbonate forms, giving a bitter taste.	1+1+1
	OR: pH decreases (lactic acid forms). Baking soda makes milk slightly alkaline so it takes longer to set into curd.	1+2

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
24	(a) decomposition (b) displacement (c) double displacement (precipitation) (d) combination (e) redox (ZnO reduced, C oxidised)	1+1+1+1+1

	OR: (i) Slow destruction of metal by surroundings (ii) Nails in: A — water + air (rusts); B — boiled water with oil layer (no air, no rust); C — dry air with anhydrous $CaCl_2$ (no rust) (iii) galvanising/painting	1+3+1
25	(i) $2\,HgS + 3\,O_2 \longrightarrow 2\,HgO + 2\,SO_2$; $2\,HgO \longrightarrow 2\,Hg + O_2$. $2\,Cu_2S + 3\,O_2 \longrightarrow 2\,Cu_2O + 2\,SO_2$; $2\,Cu_2O + Cu_2S \longrightarrow 6\,Cu + SO_2$ (ii) they are low in the reactivity series	4+1
	OR: (i) purifying an impure metal by electrolysis (ii) impure Cu as anode, pure Cu strip as cathode in acidified $CuSO_4$ solution (iii) anode — impure Cu, cathode — pure Cu, electrolyte — acidified $CuSO_4$ (iv) below the anode	1+2+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET C

Chapters 1–4: **Chemical Reactions and Equations, Acids, Bases and Salts, Metals and Non-metals, Carbon and its Compounds** | Class 10 Science | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 12 × 1 = 12

Q.	Correct Option	Brief Reasoning
1	**(a)** Student A is correct	Zn displaces Cu
2	**(a)** MgO	$2\,Mg + O_2 \longrightarrow 2\,MgO$
3	**(b)** 2 : 1	$2\,H_2O \longrightarrow 2\,H_2 + O_2$
4	**(b)** hydrogen	$2\,NaOH + Zn \longrightarrow Na_2ZnO_2 + H_2$
5	**(a)** highly exothermic	So acid is added to water slowly
6	**(a)** 10	Basic: pH > 7
7	**(d)** silver sulphide	H_2S in air
8	**(c)** Incorrect — mercury is a liquid metal	Mercury
9	**(d)** 3 : 1	Dissolves gold
10	**(c)** ethanoic acid	Oxidation
11	**(c)** catenation	Self-linking
12	**(d)** acetic acid	Vinegar is 5–8% solution

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
13	$H_2 + Cl_2 \longrightarrow 2\,HCl$ — combination reaction	1+1
14	They contain acids which react with the metal to form toxic compounds, spoiling the food.	2
15	Metal: mercury; non-metal: bromine	1+1
16	Clusters of soap molecules in water with hydrophobic tails inside and ionic heads outside, trapping oily dirt.	2
17	C_2H_5OH (ethanol) and C_3H_7OH (propanol)	1+1

SECTION — C (Short Answer) — Answers — 6 × 3 = 18

Q.	Expected Answer / Value Points	Marks
18	(i) Hydrogen at cathode, oxygen at anode (ii) 2 : 1 (iii) to increase the conductivity of water	1+1+1
19	Fixed number of water molecules in a crystal. $CuSO_4 \cdot 5\,H_2O$ loses water on heating and becomes white anhydrous $CuSO_4$; turns blue again with water.	1+1+1
20	Roasting: heating sulphide ore in air, $2\,ZnS + 3\,O_2 \longrightarrow 2\,ZnO + 2\,SO_2$. Calcination: heating carbonate ore in limited air, $ZnCO_3 \longrightarrow ZnO + CO_2$.	1½+1½
21	(i) $CH_3CH_2CH_2CH_2CH_2Br$ (ii) $CH_3COCH_2CH_3$ (iii) $CH_3(CH_2)_4CHO$	1+1+1
22	$NaCl + H_2O + CO_2 + NH_3 \longrightarrow NH_4Cl + NaHCO_3$; uses: baking powder, antacid, soda-acid fire extinguishers	1+1+1
	OR: Acid rain lowers the pH of river water; aquatic animals survive only in a narrow pH range (about 7–7.8) close to neutral.	2+1
23	(i) ions become free to move (ii) strong inter-ionic attraction needs a lot of energy (iii) water is a polar solvent; kerosene is non-polar	1+1+1
	OR: (i) $2\,Na + 2\,H_2O \longrightarrow 2\,NaOH + H_2$ (catches fire) (ii) $Fe + CuSO_4 \longrightarrow FeSO_4 + Cu$ (iii) $2\,Cu + O_2 \longrightarrow 2\,CuO$	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
24	(i) Zinc granules + dilute H_2SO_4 in a test tube; gas passed through soap solution forms bubbles (ii) $Zn + 2\,HCl \longrightarrow ZnCl_2 + H_2$ (iii) burning candle near bubble — pop sound	3+1+1
	OR: (i) Strong: ionise completely (HCl); weak: partly (CH_3COOH) (ii) mixture of indicators showing different colours at different pH (iii) (a) stomach HCl; excess acid neutralised by antacids (b) plants need a specific pH; acidic soil treated with lime	2+1+2

25	(i) $\mathrm{CH_3COOH + C_2H_5OH \xrightarrow{acid} CH_3COOC_2H_5 + H_2O}$ (ii) alkaline hydrolysis of ester to give alcohol and sodium salt of acid: $\mathrm{CH_3COOC_2H_5 + NaOH \longrightarrow C_2H_5OH + CH_3COONa}$ (iii) perfumes/flavouring	2+2+1
	OR: (i) catenation — forms long chains, branches, rings; tetravalency — bonds with 4 atoms incl. O, N, S, Cl (ii) $\mathrm{CH_3-C(=O)-OH}$ (iii) aldehyde (–CHO); ketone (C=O)	2+1+2

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